

Example: Liouville Equation and Tensor Product Identity

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Problem

Show that

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \otimes \hat{A} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes \hat{B} = \hat{A} \oplus \hat{B}$$

where $\otimes$ denotes the tensor product and $\oplus$ denotes the direct sum of operators.

1 Direct Sum of Operators

For two operators $\hat{A}$ and $\hat{B}$, their **direct sum** is defined as

$$\hat{A} \oplus \hat{B} = \begin{pmatrix} \hat{A} & 0 \\ 0 & \hat{B} \end{pmatrix}$$

This operator acts on the combined Hilbert space

$$\mathcal{H} = \mathcal{H}_A \oplus \mathcal{H}_B$$

The direct sum therefore creates a block-diagonal operator.

2 Tensor Product Structure

Consider the matrices

$$P_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$P_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

These matrices act as **projectors** selecting the first and second components respectively.

3 Tensor Product with Operator $\hat{A}$

The tensor product

$$P_1 \otimes \hat{A}$$

produces a block matrix.

Using the tensor product rule

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \otimes M = \begin{pmatrix} aM & bM \\ cM & dM \end{pmatrix}$$

we obtain

$$P_1 \otimes \hat{A} = \begin{pmatrix} 1\hat{A} & 0\hat{A} \\ 0\hat{A} & 0\hat{A} \end{pmatrix}$$

which simplifies to

$$P_1 \otimes \hat{A} = \begin{pmatrix} \hat{A} & 0 \\ 0 & 0 \end{pmatrix}$$

4 Tensor Product with Operator $\hat{B}$

Similarly,

$$P_2 \otimes \hat{B}$$

gives

$$P_2 \otimes \hat{B} = \begin{pmatrix} 0\hat{B} & 0\hat{B} \\ 0\hat{B} & 1\hat{B} \end{pmatrix}$$

which simplifies to

$$P_2 \otimes \hat{B} = \begin{pmatrix} 0 & 0 \\ 0 & \hat{B} \end{pmatrix}$$

5 Summing the Two Operators

Now add the two matrices:

$$\begin{aligned} & (P_1 \otimes \hat{A}) + (P_2 \otimes \hat{B}) \\ &= \begin{pmatrix} \hat{A} & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \hat{B} \end{pmatrix} \\ &= \begin{pmatrix} \hat{A} & 0 \\ 0 & \hat{B} \end{pmatrix} \end{aligned}$$

But this is precisely the definition of the direct sum:

$$\boxed{\hat{A} \oplus \hat{B} = \begin{pmatrix} \hat{A} & 0 \\ 0 & \hat{B} \end{pmatrix}}$$

Thus we have shown that

$$\boxed{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \otimes \hat{A} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes \hat{B} = \hat{A} \oplus \hat{B}}$$

6 Physical Interpretation

This identity has important meaning in quantum mechanics and quantum information theory.

- The matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

act as **projectors** selecting different sectors of the Hilbert space.

- The tensor product embeds operators into a larger space.
- The resulting operator becomes block diagonal, which corresponds to a direct sum structure.
- This type of structure appears frequently in
 - Liouville space representations
 - open quantum systems
 - quantum channels
 - block Hamiltonians

7 Connection to the Liouville Equation

In quantum statistical mechanics, the Liouville–von Neumann equation is

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho]$$

When expressed in vectorized (Liouville) space, operators acting from the left and right can be represented using tensor products such as

$$H \otimes I \quad \text{and} \quad I \otimes H^T$$

The identity proven above helps construct block operators used in such formulations.

8 Conclusion

We have shown that

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \otimes A + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes B$$

produces the block matrix

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$$

which is exactly the **direct sum** $A \oplus B$.

This identity illustrates how tensor products can generate block-diagonal operators, a structure widely used in quantum mechanics and Liouville-space formulations.